Khair Ul Wara

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Lahore, Pakistan

Professional Summary

Biomedical Engineering researcher (CGPA 3.89/4.00) specializing in the convergence of embedded AI, stochastic signal processing, and volumetric computer vision. Proven expertise in architecting low-latency physiological sensing systems, notably the UPEC multi-modal wearable, which successfully deployed hybrid CNN-Transformer and CNN-LSTM models on ESP32 hardware to achieve 92.3% ECG classification accuracy with <5-second latency. Currently pioneering high-fidelity 3D reconstruction pipelines using 3D Gaussian Splatting, SAM 3D, and COLMAP to establish kinematic ground truths for EMG-torque calibration and haptic feedback. A published author in the *Chemical Engineering Journal* (2025) and recipient of the Microsoft Imagine Cup award for IoT clinical tools, I am committed to advancing the next generation of unobtrusive, intelligent diagnostics and "smart" wireless sensors, personalized cardiovascular monitoring, and rehabilitative medicine.

Education

B.Sc. Biomedical Engineering

Riphah International University, Lahore, Pakistan *2021 – 2025*

CGPA: 3.89 / 4.00

Research Experience

Research Assistant, BioEra Lab, Karachi *November 2023 – January 2024*

- Developed and calibrated wearable sensors for non-invasive glucose and cardiovascular monitoring.
- Preprocessed and analyzed multimodal signals (ECG, PPG, PCG) for AI-based predictive models.
- Contributed to manuscript drafting and statistical analysis.

Independent Research Projects

- *UPEC: Multi-Modal Wearable Device for Physiological Monitoring*

  Led the development of a wearable system that simultaneously tracks ECG, PCG, and PPG signals using ESP32. Built hybrid CNN-LSTM and Transformer models that achieved over 92% accuracy in real-time on the device itself. The project taught me how important low latency and

comfortable design are when people need to wear something every day, an experience I believe will be valuable when designing tools for young people managing ADHD symptoms.

- *Advanced 3D Computer Vision for Movement Analysis & Haptics (Ongoing)*

  Developing high-fidelity 3D reconstruction and segmentation pipelines using Segment Anything Model 3D (SAM 3D), 3D Gaussian Splatting, and COLMAP. These tools enable precise tracking of human movement and anatomical structures directly applicable to haptic feedback, tangible interfaces, and tele-tangible systems for rehabilitation.

- *Real-Time Patient Monitoring System*

  Developed a portable IoT platform (PPG, SpO2, temperature, CO2) with cloud connectivity that earned 2nd Runner-up at Microsoft Imagine Cup Pakistan. This project made me realize how powerful continuous, unobtrusive monitoring can be for remote care. I'm eager to explore similar approaches for tracking attention patterns, activity levels, or sleep quality in neurodiverse youth.

- *AI for Physiological Signal Analysis*

  Worked extensively with ECG, PPG, and PCG data from preprocessing to building predictive models. I've become particularly interested in how small physiological changes can reflect bigger shifts in focus or emotional state, which feels highly relevant to ADHD research.

- Conducted biomedical imaging research using Vision Transformer (ViT) for neurodegenerative disease detection.

Publications

- Bangul Khan, Rana Talha Khalid, **Khair Ul Wara**, Muhammad Hasan Masrur, et al. Reshaping the Healthcare World by AI-Integrated Wearable Sensors Following COVID-19. *Chemical Engineering Journal*, Vol. 505, 2025, Article 159478.

- **Khair Ul Wara**, Muhammad Hasan Masrur, et al. Advancements in Ultrasound Gel Pad Technologies: Enhancing Diagnostic Precision, Procedural Efficiency, and Therapeutic Applications. (Under Review)

- Muhammad Hasan Masrur, Rana Talha Khalid, **Khair Ul Wara**, et al. A Semi-Automated Framework for Standardized Vertebral Measurement with Enhanced Reproducibility in Lumbar Spine MRI Analysis. *Materials Proceedings*, 2025.

- Muhammad Daniyal Maqsood Alvi, **Khair Ul Wara**, et al. Evaluating Machine Learning

Classifiers for Automated Glaucoma Detection Using Fundus Images. *Biomedical Materials & Devices*, 2025.

- Muhammad Daniyal Maqsood Alvi, **Khair Ul Wara**, et al. IoT-enabled Non-invasive Glucose Monitoring and Heart Rate Detection. *Innovative Computing Review*, 2025.

Technical Research Competencies

- AI & Machine Learning: CNN, LSTM, Vision Transformers (ViT), multimodal fusion, time-series analysis.
- Wearable & Embedded Systems: ESP32, real-time signal acquisition, low-power optimization, sensor integration.
- Signal Processing: ECG, PPG, PCG filtering, segmentation, spectral analysis, wavelets.
- Biomedical Imaging: MRI preprocessing, feature extraction, ViT-based classification.
- 3D Computer Vision: Segment Anything Model 3D (SAM 3D), 3D Gaussian Splatting, COLMAP structure-from-motion.
- Neuroimaging & Biomechanics: MRI preprocessing, Vision Transformers, 3D motion reconstruction for rehabilitation.

Selected Research Projects

- UPEC: Multi-Modal AI-Powered ECG/PCG/PPG Device – Developed CNN-Transformer and CNN-LSTM hybrid models achieving 92.3% ECG classification accuracy and 92.1% murmur detection F1-score; deployed real-time inference on ESP32 (<5-second latency).
- Real-Time Patient Monitoring System (IoT-based) – Microsoft Imagine Cup Pakistan 2nd Runner-up.
- Non-Invasive Glucose Monitoring (breath-acetone sensing prototype).
- Alzheimer's Detection using Vision Transformer (ViT) on MRI images.
- Neuroimaging & Motor Recovery Monitoring (ongoing 3D vision work with SAM 3D, Gaussian Splatting, COLMAP).

Industry Experience

- **Intern**, Vertex Medical Pvt, Lahore *2023*
- **Intern**, EndoKare Pvt Ltd, Lahore *2024*

Leadership & Volunteering

President, Riphah Society of Engineering & Sciences (RSES)

Volunteer: Amnesty International, Crisis Text Line, Movers Program

Awards & Honors

- Talent Scholarship (2021, 2023)
- 2nd Runner-up, Microsoft Imagine Cup Pakistan
- Mumtaz Scholarship (2023, 2024)